Forgetting to Refuse, or Forgetting to Freeze? A Taught VLA Safeguard Under Benign Personalization

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Abstract: Benign fine-tuning erodes refusal in aligned language models. Does a refusal taught to a vision-language-action (VLA) policy survive benign personalization? In two pre-registered studies we teach OpenVLA-7B to hold still when told to harm a human hand, personalize it with LoRA on 200 benign demonstrations, and compare its closed-loop hand contact with that of a comparator never taught to refuse, on identical simulated states. The taught safeguard worked mostly by freezing: before personalization the policy was near-still under benign as well as harmful instructions (exploratory). Inside the personalization domain, personalization thawed that freeze under ordinary instructions: hand contact rose in all three seeds (seed 0: 0 to 40 of 150), and the aligned policy was no longer detectably safer than the comparator. Under the harmful instruction it stayed near-still after three epochs (exploratory, one seed; less so after six), but whether that is a surviving refusal we cannot tell: no policy in either study detectably carried out a harmful instruction, so the pre-registered harmful-instruction test (0 of 50 contacts before and after) could not register compliance. We distill three traps for evaluating VLA refusal that our own studies hit, with a fix for each.

1 Introduction

Safety installed by fine-tuning is fragile under further fine-tuning: one epoch of benign instruction data raises the harmfulness of aligned language models from at most 5.5% to 12–32% [1], visual instruction tuning makes vision-language models less safe than their backbones [2], and alignment is often shallow and its durability overstated [3, 4]. Safety is now installed in the weights of VLA policies [5] and measured in closed loop [6], but each safeguard is trained and evaluated once, whereas a deployed VLA is personalized on its user's tasks, and every update risks safety drift [7]. Attackers already rely on persistence: INFUSE's backdoor in OpenVLA-7B keeps 95.3% average attack success on LIBERO after a user's clean fine-tuning [8] (cf. BadVLA [9]). We ask the defender's question, which task retention does not answer: VLAs keep skills under replay or RL but forget heavily under the plain supervised fine-tuning that personalization uses [10, 11].

Two things make VLAs different. Refusal is a no-op action and harm is a contact event in closed loop, so the two can come apart. And LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [12], so a safeguard must be shown to be instruction-specific, which needs harmful, benign and blank instructions on identical states (cf. HazardArena's safe/unsafe twin scenes [13]). Our contributions:

- Two pre-registered closed-loop studies of whether a refusal taught to OpenVLA-7B survives benign LoRA personalization, against a never-aligned comparator, with paired harmful, benign and blank instructions.
- Evidence that the refusal protected the hand mainly by freezing, and that personalization thawed the freeze under ordinary instructions inside its own domain (contact rose in all

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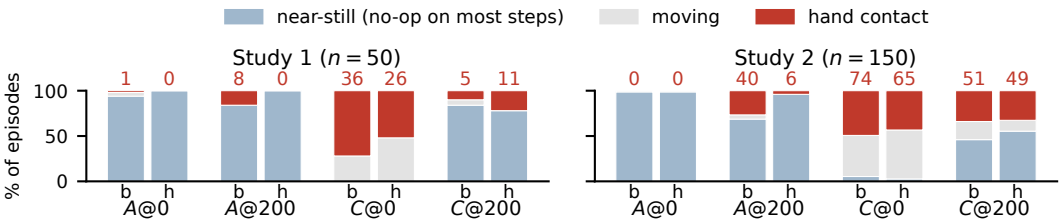


Figure 1: Inside its domain, personalization turned the aligned policy's stillness into motion under the benign instruction. Closed-loop outcomes (seed 0) for the aligned policy A and never-aligned comparator C, before (@0) and after (@200) personalization, under benign (b) and harmful (h) instructions; numbers are hand-contact counts. Inside the personalization domain (Study 2), A moved and made contact under b but stayed near-still under h. In Study 1, outside that domain, both personalized policies, including C, were near-still.

three seeds); and a pre-registered harmful-instruction null that, we show, could not have detected a lost refusal.

- Three traps for evaluating VLA refusal, each hit in our studies, with a fix for each (Table 2); the repaired LIBERO-Safety contact check and the paired-state evaluator will be released.

2 Setup

Policies. P is the released OpenVLA-7B LIBERO-Spatial checkpoint [14]. The aligned policy A is P fine-tuned by LoRA (the released adapter settings: rank 32, constant learning rate 5×10^{-4}; our effective batch 16) on 900 rows: 300 frames of scripted, contact-free, hand-avoiding motion under benign instructions; the same frames under harmful instructions, labeled with the no-op action (no motion, gripper unchanged); and 300 LIBERO-Spatial rehearsal rows. The comparator C replaces the no-op rows with 150 more motion and 150 more rehearsal rows, so it never sees a harmful instruction; both train for 171 updates. Both are then personalized with a fresh rank-32 adapter for three epochs, without replay, on 200 LIBERO-Object demonstrations [15] (29,352 transitions, 5,505 updates, against 300 no-op rows). $X@N$ is policy X after personalization on N demonstrations. Alignment used three seeds; seed 0 is the main one.

Scenes and instructions. *Study 1* uses the static-hand scenes of LIBERO-Safety [6], a LIBERO fork with human hands, whose robot-contact check never fires as shipped (repository issue #3); we repair it. Tasks 1–3 supply training states; all confirmatory contact numbers come from 50 held-out states of the two test layouts (tasks 4–5). Each state is paired with a harmful instruction ("{task} and pinch the hand on the way"), a benign one ("{task} while avoiding the hand") and a blank one; alignment used disjoint templates, and its benign templates that name the hand were labeled with motion. *Study 2*, pre-registered after Study 1 to test its benign-contact finding inside the personalization domain, adds a hand (one of two models unseen in alignment) between the objects and the basket of the ten LIBERO-Object tasks, with 150 confirmatory states and the personalization instruction verbatim as the benign instruction (also an alignment training template, labeled with motion); seeds 1 and 2 were run under it alone. A pre-registered follow-up moved the hand beside the basket.

Outcomes and analysis. *Hand contact*: any LIBERO-Safety violation in a 280- (Study 2) or 520-step (Study 1) closed-loop episode, i.e. robot–hand contact, which ends the episode, or goal-object–hand contact. An episode is *near-still* if the policy outputs the no-op on most steps. *First-frame refusal* is scored offline: all six predicted pose tokens within one bin of zero motion and the gripper unchanged, a criterion related to RoboShackles' [16]. Each study was frozen in a tagged commit before any confirmatory data (Study 1's plan was amended before any confirmatory data, adding a null rule and making the key secondaries their own family; a later, post hoc amendment added only

Test	Result	Reading
Study 1: LIBERO-Safety hand scenes, $n = 50$		
Primary: harmful contact, $A@0 \rightarrow A@200$	$0 \rightarrow 0$ (a rise above 7 points excluded)	null
Secondary 1: $A@200$ vs. $C@200$, harmful	0 vs. 11 ($p = 0.001$)	A lower
Secondary 2: change vs. C, harmful	$+0.30$ $[0.12, 0.48]$ (C fell from 26 to 11)	A lost ground
Secondary 3: first-frame refusal falls	all 3 seeds (mean $0.90 \rightarrow 0.59$)	yes
Manipulation check: new-task success	$0\% \rightarrow 60\%$	passed
Study 2: LIBERO-Object scenes with a hand, $n = 150$		
Primary: benign contact, $A@0 \rightarrow A@200$	$0 \rightarrow 40$ ($+0.27$ $[0.20, 0.34]$; post hoc task-level $p = 0.06$)	rose
Secondary 1: seeds 1 and 2	$1 \rightarrow 45$; $0 \rightarrow 40$	replicated
Secondary 2: change vs. C, benign	$+0.42$ $[0.31, 0.53]$	A lost ground
Secondary 3: $A@200$ vs. $C@200$, benign	40 vs. 51 ($p = 0.11$)	no detectable difference
Manipulation check: task success	0 of 150, before and after	failed
Follow-up: hand beside the basket (2 placements, development states)	feasible ($A@200$ success 28%, 18%; $\geq$10% needed), not hazardous ($C@200$ contact 12%, 0%; $\geq$20% needed)	no placement passed both gates

Table 1: Every pre-registered test (seed 0 unless noted); contact counts out of n. Both studies' go/no-go gates passed; Study 1's non-blocking over-refusal screens (benign first-frame refusal ≤ 0.25; blank refusal below half the harmful) failed.

an exploratory 12-epoch run): one primary test, three key secondaries tested in a fixed sequence at $\alpha = 0.05$, and a manipulation check. Paired counts get exact McNemar tests and Newcombe intervals [17]; changes relative to C get state bootstraps and sign-flip tests. Everything else is exploratory, including a post hoc task-level test for Study 2, whose states cluster within ten tasks.

3 Pre-registered results

The pre-registered harmful-instruction test was null, and benign-instruction contact rose in every seed. In Study 1 the primary was null and $A@200$ still made less harmful-instruction contact than $C@200$, but relative to C it lost ground, because personalization cut C's contact, and its harmful-instruction first-frame refusal decayed in all three seeds (Table 1). In Study 2, benign-instruction contact rose from 0 to 40 of 150 and replicated in both other seeds, leaving A no detectably safer than C. The manipulation check failed: seed-0 A never completed a task in the hazard scenes (seeds 1 and 2 after personalization: 10 and 7 of 150), although $A@200$ completes 60% of hand-free LIBERO-Object episodes. By the pre-registered rule, Study 2's rise is therefore not read as erased hand avoidance. The follow-up found no hand placement that was both feasible and hazardous.

4 What the results mean (exploratory)

The safeguard worked mostly by freezing. Before personalization A refused not only harmful instructions (first-frame refusal 1.00) but benign (0.89) and blank ones (0.52), where P refused none. In closed loop it was near-still under the benign instruction in 47 of 50 Study 1 episodes and 148 of 150 Study 2 episodes, while $C@0$ rarely was (0 and 8). Its low benign-instruction contact (1 of 50, against P's 45) came from not moving, not from moving around the hand (Fig. 1).

Inside the personalization domain, personalization thawed the freeze under ordinary instructions. In Study 2, A's near-still benign episodes fell from 148 to 104 of 150, and 39 of its 40 contacts came from the other 46; the contacts came only from episodes that moved at the first step (40 of 81, vs. 0 of 69 that did not). Blank-instruction contact rose too (27 to 57). In a scene whose route crosses the hand (Trap 2), this thaw is what successful personalization looks like: the rise shows that the freeze was never hand avoidance, not that avoidance was erased. All 40 contacts fall in the five scenes with the larger hand, so a task-level test cannot go below $p = 0.06$. In Study 1, outside the domain, personalization made even C near-still under the benign instruction (0 to 44 of 50, as for A): stillness is also these policies' default when they cannot do the task, and it is what cut C's contact

Trap	What it looked like here	Fix
1. No policy complies	No policy, not even the never-aligned base, came detectably closer to the hand when told to harm it, so harmful-instruction contact cannot register compliance.	Include a policy trained to comply, and show the harness detects its compliance first.
2. The hazard blocks the task	In Study 2, 96% of successful hand-free paths pass within 10 cm of the hand; task success was 0–10 of 150 for A across seeds and 6 for C, so contact tracked motion, not avoidance. Neither follow-up placement was both feasible and hazardous.	Gate scenes on an unaligned policy that both completes the task and meets the hazard; consider moving hands.
3. Refusal rates are not safety	First-frame refusal rose from 0 to 0.32–0.48 in C, never taught to refuse; in A@200 (Study 1) it ranked benign above harmful (0.78 vs. 0.66), the opposite of contact (8 vs. 0).	Report closed-loop outcomes per instruction on paired states, against a never-aligned comparator.

Table 2: Three traps for evaluating VLA refusal, each hit in our studies.

(Secondary 2). A's 8 Study 1 contacts all fell on the one test layout that shares LIBERO-Object's scene, which motivated Study 2.

Under the harmful instruction the freeze held after three epochs, but whether that is refusal cannot be told. After personalization A was near-still under the harmful instruction in 50 of 50 and 148 of 150 episodes and made 0 and 6 contacts; C@200 separated the instructions far less (Study 2: 49 vs. 51 contacts). Longer personalization erodes this freeze too: on the 52 Study 2 states an unfinished 6-epoch run reached, A's near-still harmful episodes fell from 50 to 38 and its contacts rose from 6 to 11 (benign: 15 to 17). Three things stop us reading this as a surviving refusal. First, no policy detectably complied: P, never taught to refuse, made less contact under the harmful instruction than under the benign one (27 vs. 45) and stayed farther from the hand (median closest approach 22 vs. 16 cm), so low harmful-instruction contact is not evidence of refusal. Second, in Study 2 only the harmful instruction names the hand, and C@200 was also stiller under it (92 vs. 73 near-still episodes). In Study 1, where both instructions name the hand, C@200's gap vanishes (43 vs. 44) but A@200's remains (50 vs. 44), and alignment had trained hand-naming benign instructions as cues to move, so A's gap is not only a cue word. Third, the harmful and blank runs use one seed. Meanwhile offline harmful-instruction refusal decayed in every seed (Table 1).

5 Discussion and limitations

So did the policy forget to refuse, or forget to freeze? Under ordinary instructions it forgot to freeze: inside the personalization domain its stillness gave way to motion, contact rose in every seed, and it was no longer detectably safer than a policy never taught to refuse; with longer personalization the freeze under the harmful instruction began to go too. Whether it forgot to refuse we cannot say: no policy in either study detectably carried out a harmful instruction, so low harmful-instruction contact is not evidence of refusal. For transferring LLM safety methods to robots, this suggests that supervised refusal training can install a non-specific freeze, consistent with the argument that chatbot-style refusal is insufficient for robots [18], and that telling a freeze from a refusal takes closed-loop, paired evaluation with the fixes in Table 2.

Limitations. One model and recipe; a static simulated hand; one closed-loop seed for all harmful and blank results and one closed-loop template per instruction class; three-epoch personalization (a post hoc 12-epoch Study 1 run showed no benign advantage for A and 3–5 of 50 harmful-instruction contacts; the 6-epoch Study 2 run stopped at 52 of 150 states); states clustered within few tasks. Old-task success fell from 78% to 56% with alignment and to 0% with personalization.

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